

Programming SQL Server 2019

Database Terms and Concepts and Introducing SQL Server

Topics

- Database & RDBMS
- SQL Server
- SQL/T-SQL
- SQL Server authentication
- Introducing
- SQL Server and Services installed
- Various Tools
- Management tools
- Configuration
- SQL Server instances
- Connecting to SQL Server.

Important Points:

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Summary

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Terms and Concept

What is a database?

- A database is a collection of information that is organized so that it can easily be accessed, managed, and updated.
- In computing, databases are sometimes classified according to their organizational approach.
- The most prevalent approach is the relational database, a tabular database in which data is defined so that it can be reorganized and accessed in a number of different ways.

What is relational database?

- A relational database is a collection of data items organized as a set of formally described tables from which data can be accessed easily.
- The standard language to communicate a relational database is the structured query language (SQL).

What is Relational Database Management System (RDBMS)?

- A relational database management system (RDBMS) is a program that lets you create, update, and administer a relational database.
- Most commercial RDBMS's use the Structured Query Language (SQL) to access the database, although SQL was invented after the development of the relational model and is not necessary for its use.
- The leading RDBMS products are Oracle, IBM's DB2 and Microsoft's SQL Server.

What is SQL?

- SQL (Structured Query Language) is a special-purpose programming language designed for managing data in relational database management systems (RDBMS).
- Structured Query Language (SQL) is a specialized language for updating, deleting, and requesting information from databases. SQL is an ANSI and ISO standard.
- A variety of established database products support SQL, including products from Oracle and Microsoft SQL Server.

What is Client/Server computing?

- The client/server model is a computing model that acts as a distributed application which partitions tasks or workloads between the providers of a resource or service, called servers, and service requesters, called clients.
- Often clients and servers communicate over a computer network on separate hardware, but both client and server may reside in the same system.

- A server machine is a host that is running one or more server programs which share their resources with clients.
- A client does not share any of its resources, but requests a server's content or service function. Clients therefore initiate communication sessions with servers which await incoming requests.

What is database server?

- Database server is the term used to refer to the back-end system of a database application using client/server architecture.
- The back-end, sometimes called a database server, performs tasks such as data analysis, storage, data manipulation, archiving, and other non-user specific tasks.

What is SQL server?

- Microsoft SQL Server is a relational database management system developed by Microsoft.
- As a database server, it is a software product whose primary function is to store and retrieve data as requested by other software applications - which may be running on the same computer or on another computer across a network (including the Internet).
- There are at least a dozen different editions of Microsoft SQL Server aimed at different audiences and for different workloads (ranging from small applications that store and retrieve data on the same computer, to millions of users and computers that access huge amounts of data from the Internet at the same time).

What is T-SQL?

- Transact-SQL (T-SQL) is Microsoft's and Sybase's proprietary extension to SQL.
- Transact-SQL is central to using Microsoft SQL Server. All applications that communicate with an instance of SQL Server do so by sending Transact-SQL statements to the server.

What is a SQL Server instance?

- An SQL Server instance is a complete SQL server.
- Each complete installation of SQL server is an instance and each is identified by a unique name.
- You can install many instances on a machine but you can have only one default instance.
- On a given computer you can run multiple SQL Server services, each with their own ports, logins, and databases. Each of these services is called an "instance".

What is SQL Server Management Studio?

- SQL Server Management Studio (ssms) is a software application first launched with the Microsoft SQL Server 2005 that is used for configuring, managing, and administering all components within Microsoft SQL Server.

- The tool includes both script editors and graphical tools which work with objects and features of the server.

What is SQL Server Configuration Manager?

- SQL Server Configuration Manager is a tool to manage the services associated with SQL Server, to configure the network protocols used by SQL Server, and to manage the network connectivity configuration from SQL Server client computers.

What is SQL server user and login?

- Although the terms login and user are often used interchangeably, they are very different.
- A login is used for user authentication
- A database user account is used for database access and permissions validation.
- Logins are associated to users by the security identifier (SID).
- A login is required for access to the SQL Server.
- A login must be associated to a SQL Server database user.
- You use the user account to control activities performed in the database.

What is authentication?

- The process of verifying that a particular login is valid is called "authentication".

What are SQL server authentication modes?

- Microsoft® SQL Server™ can operate in one of two security (authentication) modes:
 - Windows Authentication Mode (Windows Authentication)
- Windows Authentication mode allows a user to connect through a Windows user account.
- Mixed Mode
- Allows using Windows Authentication and SQL Server Authentication.

What is the difference between Windows Authentication and SQL Authentication?

- Windows Authentication
- When a user connects through a Windows user account, SQL Server revalidates the account name and password by calling back to Windows for the information.
- In this case SQL Server does not store login information own its own and depends windows for authentication.
- If you use windows authentication SQL server does not ask you to provide login and password.
- SQL Server authentication
- In SQL Server authentication, SQL server stores login information of pre-created users.
- A user connects with a specified login name and password from a , SQL Server performs the authentication itself by checking to see if a SQL Server login account has been set up and if the specified password matches the one previously

recorded. If SQL Server does not have a login account set, authentication fails and the user receives an error message.

Introduction to the SQL Server

Evolution of SQL Server

- SQL server is the product of Microsoft and Sybase. That is the reason SQL server share some common features in earlier releases.
 - After some year Microsoft went separate and introduced SQL Server 2000.
 - After some long years Microsoft released new SQL server 2005.
- It was renovated, new feature added and .NET.
- At lot new services were added
- New tools were introduced
- New configuration system was included
- Microsoft continued to bring enhancements, improvements, adding new features
 - Then SQL Server 2008 and SQL 2008 R2 were released.
 - Then came SQL Server 2012 in market with Cloud-based database support.
 - Currently SQL Server 2014 in market

SQL server Editions

- SQL server come in number of editions depending on need and supports.
- Enterprise editions - the highest editions with all features and most costly one.
- There are professional, standard edition too - you have to buy license to use.
- But Microsoft provide developers a free edition called express editions.
- Express editions come with some features and services curtailed but all basic and necessary features required for development included.
- We will use SQL server 2012 express.

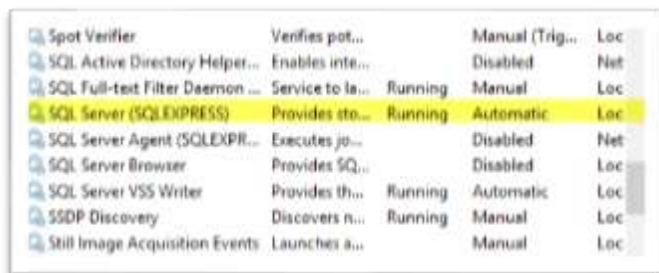
Components and services installed with SQL Server

- With SQL server a lot of things are installed:
- Number of services
- Configuration tools
- Management tools
- Query development tools

Where is SQL database server?

- It is installed as windows service - waiting for client.
- [A windows service is a special type of software that runs in background. Windows services do not require users to run it, it can start automatically when windows starts. They have no user interface but can be controlled using Windows Service Control Manager]
- It can be found inside Windows service manager.
- Let's see it
- Press windows button+ R

- Type Services.msc and press enter
- Service Control Manager will open
- Search from SQL Server
- In your lab machines, you will see like what you see in figure 1



Spot Verifier	Verifies pot...	Manual (Trig...	Loc
SQL Active Directory Helper...	Enables inte...	Disabled	Net
SQL Full-text Filter Daemon ...	Service to la...	Running	Manual
SQL Server (SQLEXPRESS)	Provides sto...	Running	Automatic
SQL Server Agent (SQLEXPRESS)	Executes jo...	Disabled	Net
SQL Server Browser	Provides SQ...	Disabled	Loc
SQL Server VSS Writer	Provides th...	Running	Automatic
SSDP Discovery	Discovers n...	Running	Manual
Still Image Acquisition Events	Launcher a...	Manual	Loc

SQL Server instance

- You can have multiple SQL Server in the same computer
- Each complete install is known as instance
- Each must have a name but one can be installed without name - called default instance
- To communicate with a particular instance you must know its name
- Each instance name is associated with the computer name on which it is installed.
- Suppose your computer name ServerPC and you have three instances:
 - One default instance
 - Two named instances SQLSrvA and SQLSrvB
- Default instance is identified by the same name as the computer
- A Named instance is identified by computer -name\instance-name
 - ServerPC will refer to the default instance
 - ServerPC\SQLSrvA will refer to the instance SQLSrvA
 - ServerPC\SQLSrvB will refer to the instance SQLSrvB
- You can use . (dot) for local computer - the computer you are working on.

SQL

- To communicate with SQL server and create object (such as table) and manipulate data, you use SQL (T-SQL)
- But you can do almost all task visually using tools provided with SQL Server

Where we use SQL and How to learn and test SQL

- Generally application programs developed using languages like C, C# use SQL to manipulate data in a SQL Server
- But we have to learn SQL first and test our SQL statements
- SQL server provided some tools to write and to executes SQL statements
- There is a command line tool - SQLCMD

- There is a GUI based tool - SQL Server Management studio (SSMS), which has a Query Editor to write and test SQL
- Mostly we will use SSMS

First look of SSMS

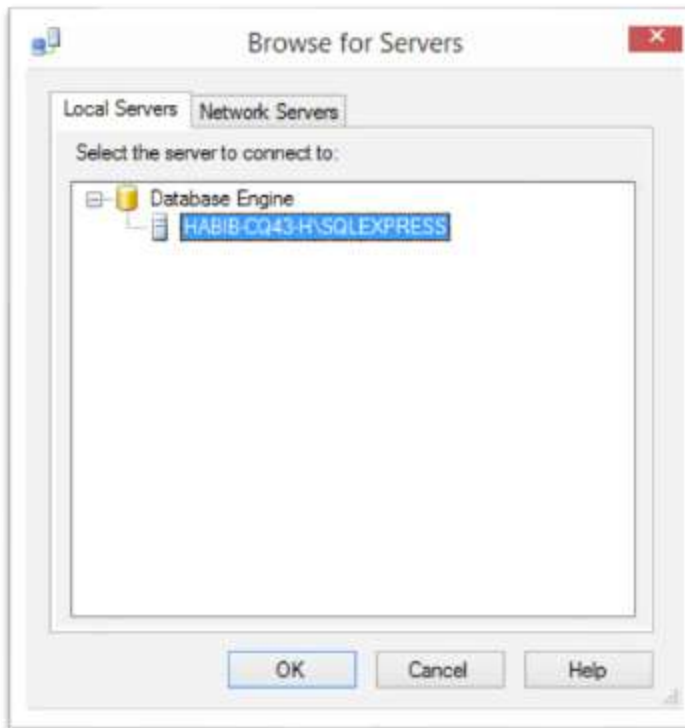
- Go to all programs > Microsoft SQL Server 2012
- Click on SQL Server Management Studio
- You will see like in figure



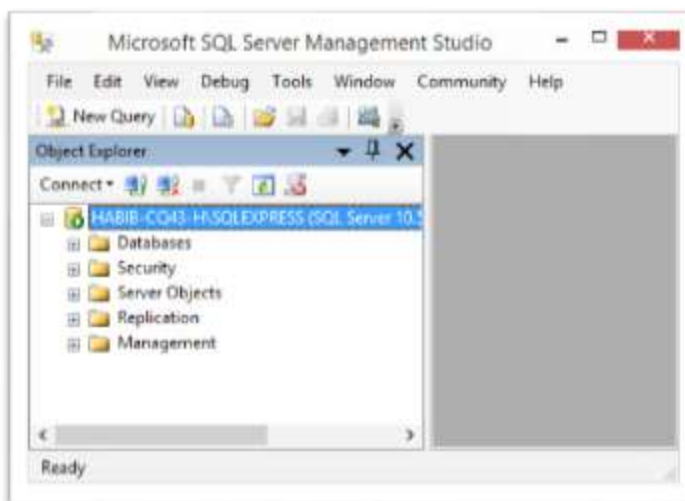
- If you have Sever name blank
- Click drop arrow next to it
- Click <Browse for more>



- Click + button on the database engine under local servers tab
- Select the server you want to connect



- Click connect button
- You will see like figure



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